Bayesian Statistics: Models, Inference, and Computation

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Lecture 3 of 3

Statistical inference: Frequentist vs Bayesian

Frequentist inference

- A random variable Y is randomly drawn through a probability model $f(y \mid \theta)$ with indexing parameter θ, which is fixed.
- A sample $\boldsymbol{y} = (y_1, \ldots, y_n)$ is observed and used to estimate the unknown value of θ, using the likelihood $L(\theta \mid \boldsymbol{y})$, allowing also to assess the sampling uncertainty.

Bayesian inference.

- The parameter θ is also a random variable, with probability distribution $\pi(\theta)$, chosen on the basis of previous studies, expert knowledge and the problem at hand.
- The likelihood function is used to update $\pi(\theta)$ and obtain the posterior distribution: $\pi(\theta \mid \boldsymbol{y}) \propto L(\theta \mid \boldsymbol{y})\pi(\theta)$.
- The posterior is used to characterize the uncertainty about θ, make prediction for future observations and model checking.

Learning objectives

By the end of the lecture, students should be able to:

- Formulate a likelihood and prior, and interpret the posterior and its summaries;
- Derive and report a one-parameter Beta–Binomial analysis;
- Explain joint and marginal posteriors in a multiparameter model;
- Computational aspects of the Bayesian statistical analysis and MCMC methods: Metropolis-Hastings, Gibbs, Hamiltonian MonteCarlo.

Application: estimating an admission probability

Consider women applying to the Berkeley graduate department coded A in fall 1973.
Let Y denote the indicator function equal to 1 in case of admission and 0 otherwise. Hence, $Y \sim \text{Bernoulli}\,(p)$, where p is the probability of admission.

- **Data model:** $Y \mid p \sim \text{Bernoulli}(p)$; the data will be the observed indicator y. The sum of the indicator variables over a sample of size n, denoted by z is thus the realization from a $\text{Binomial}(n, p)$ distribution.
- **Frequentist analysis of p:** p is fixed but unknown; given a sample of n observations, then estimate $\widehat{p} = z/n$.
- **Bayesian analysis:** before observing the data, assign the prior $p \sim \text{Beta}(1, 1)$; update it with the observed data $\boldsymbol{y}$ to obtain $\pi(p \mid \boldsymbol{y})$.

The uniform prior is chosen to express lack of information about p; in applied analyses we would examine whether it expresses reasonable knowledge before fitting.

Berkeley admissions: the observed table

The public R object `datasets::UCBAdmissions` gives admitted and total applications for each gender and department.

	Men		Women	
Department code	Admitted	Total	Admitted	Total
A	512	825	**89**	**108**
B	353	560	17	25
C	120	325	202	593
D	138	417	131	375
E	53	191	94	393
F	22	373	24	341
All six	1,198	2,691	557	1,835

The letters A–F are *codes in the released R table*. These are grouped counts for a total of 4,526 applications.

Source: R `datasets::UCBAdmissions` documentation; Bickel, Hammel and O'Connell (1975).

Admission of women to Department A

The highlighted cell is for women applying to the department coded A.

$$z = 89 \text{ admitted}, \qquad n = 108 \text{ applicants}, \qquad \widehat{p} = \frac{z}{n} = 0.824.$$

- Let p denote the probability of admission for this group.
- The observed proportion is 82.4%.
- Our inferential question is how uncertain we are about p after observing these counts.

How Bayesian data analysis informs the researcher?

Suppose we have counts of applications and offers in each Berkeley department. We want to understand the admission pattern and anticipate future outcomes.

1. **Describe uncertainty before seeing the counts.** Assign plausible ranges to the unknown admission probabilities; make explicit what is being assumed.
2. **Learn from the observed counts.** Update these ranges and ask how likely it is that women have lower admission odds than men applying to the same department.
3. **Check and predict.** Ask whether the fitted model reproduces admissions in each group; predict the number of offers among future applicants with a specified profile.

Practical value: the answer includes both an estimate and a measure of uncertainty, with assumptions that we can inspect and revise.

Posterior distribution and Bayes' rule

- Let $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$ be a vector of independently and identically distributed random variables, with observed values $\boldsymbol{y} = (y_1, \ldots, y_n)$, and let $\theta \in \Theta$ be the unknown parameter vector indexing the distribution of Y_i:

$$f\left(\boldsymbol{y} \mid \theta\right) = \prod_{i=1}^{n} f(y_i \mid \theta)$$

- The posterior distribution is then:

$$\pi(\theta \mid \boldsymbol{y}) = \frac{f\left(\boldsymbol{y} \mid \theta\right) \pi(\theta)}{f(\boldsymbol{y})}.$$

  where

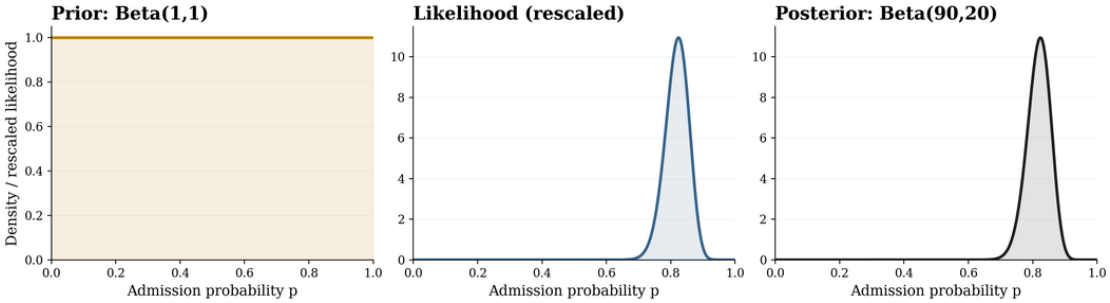
$$f(\boldsymbol{y}) = \int_{\Theta} f\left(\boldsymbol{y} \mid \theta\right) \pi\left(\theta\right) \mathrm{d}\theta$$

  is the distribution of $\boldsymbol{y}$ marginalized with respect to the "latent" θ.

- For fixed observed $\boldsymbol{y}$, $f(\boldsymbol{y})$ is the normalising constant that makes the posterior integrate (or sum if discrete) to one.

Prior, likelihood, and posterior: a preview

For the Department A example, let p be the admission probability and observe $z = 89$ offers among $n = 108$ applicants. Begin with $p \sim \text{Beta}(1, 1)$.



$$\underbrace{\pi(p \mid \boldsymbol{y})}_{\text{posterior}} \propto \underbrace{L(p \mid \boldsymbol{y})}_{\text{likelihood}} \underbrace{\pi(p)}_{\text{prior}}.$$

The likelihood is rescaled: it is not itself a probability distribution over p. With a uniform prior, its rescaled shape coincides with the posterior. We derive this result below.

Worked update: admissions in Department A

Among $n = 108$ women applying to A, $\sum_{i=1}^{108} y_i = 89$ were admitted and the remaining 19 were not. The Bernoulli likelihood and a Beta$(1, 1)$ prior give

$$\begin{aligned}\pi(p \mid \boldsymbol{y}) &\propto \underbrace{p^{89}(1-p)^{19}}_{\text{Binomial lik.}} \underbrace{p^{1-1}(1-p)^{1-1}}_{\text{Beta prior kernel}} \\ &\propto p^{89}(1-p)^{19} = p^{90-1}(1-p)^{20-1}.\end{aligned}$$

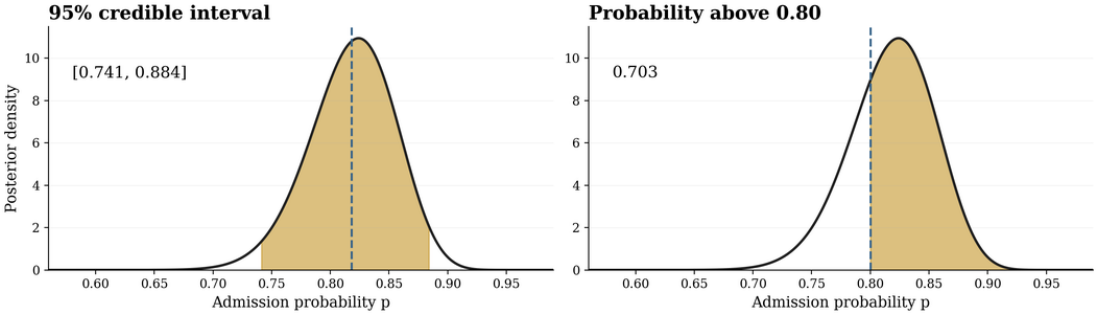
This is the kernel of a Beta density with parameters 90 and 20. Hence

$$p \mid \boldsymbol{y} \sim \text{Beta}(90, 20).$$

The Beta distribution is *conjugate* for the parameter p of a Bernoulli (and also for the Binomial) distribution. This means that the posterior distribution is of the same parametric family as the prior.

Reading a posterior distribution

The Berkeley example gives $p \mid \boldsymbol{y} \sim \text{Beta}(90, 20)$ under a uniform prior.



- The dashed line in the left panel is the posterior mean, $\mathbb{E}(p \mid y) = 0.818$; its shaded area has posterior probability 0.95.
- The right shaded area is $\mathbb{P}(p > 0.80 \mid y) = 0.703$.
- Predicting new admissions also requires allowing for Binomial variation in future outcomes.

How would we choose a prior for admissions?

For a Beta(a, b) prior on the admission probability p, $a/(a+b)$ is its mean; $a+b$ controls how concentrated it is around that mean.

- **Previous analyses:** suppose 72 of 100 applicants in a separate, comparable earlier cohort were admitted. Updating Beta$(1, 1)$ with those counts gives Beta$(73, 29)$ as a possible prior for the new cohort.
- **Expert assessment:** if an expert expects about 70% admission but is quite uncertain, Beta$(7, 3)$ has mean 0.70 and moderate concentration. These shape parameters are an elicitation.
- **Noninformative:** Beta$(1, 1)$ is uniform over $p \in [0, 1]$.

Prior-posterior sensitivity: first, inspect the number of events each prior predicts, then examine how the posterior changes under different prior distributions.

Prior predictive checks

Before updating with the observed count, simulate from the proposed model:

$$p^{(s)} \sim \underbrace{\pi(p)}_{\text{Beta}(1,1)}, \qquad Z^{(s)} \mid p^{(s)} \sim \text{Binomial}(108, p^{(s)}).$$

since $Z^{(s)} = \sum_{i=1}^{108} Y_i^{(s)}$

- Based on the sampled values of p, and thus of Z, then we need to assess whether the corresponding value of Z is plausible.
- If not, reconsider the prior before using the observed count $y = 89$.

Deriving the Beta posterior

$$\begin{aligned}\pi(p \mid \boldsymbol{y}) &\propto L(p \mid \boldsymbol{y})\, \pi(p) \\ &\propto \underbrace{p^{z}(1-p)^{n-z}}_{\text{likelihood kernel}}\ \underbrace{p^{a-1}(1-p)^{b-1}}_{\text{prior kernel}} \\ &\propto p^{a+z-1}(1-p)^{b+n-z-1}.\end{aligned}$$

Therefore,

$$p \mid z \sim \text{Beta}(a+z,\ b+n-z).$$

- The posterior remains in the Beta family. This is called conjugacy.
- The posterior parameters add the observed admissions and rejections to the prior parameters.

Convenience of conjugate priors

For a specified likelihood, a prior family is *conjugate* when the posterior belongs to the same family. In the admissions example,

$$\text{Beta}(a, b) \xrightarrow[y \text{ admissions in } n \text{ applications}]{\text{Binomial likelihood}} \text{Beta}(a + y, b + n - y).$$

- The posterior distribution has a closed form: posterior means, intervals and predictions can be algebraically calculated.
- In a multiparameter model, the *joint* posterior may not be computed in closed form. Nevertheless, there are cases where for some parameters, the *conditional* posterior distribution has a standard form. This is conditional conjugacy and can enhance the *computation of the posterior distribution*.
- Computational convenience is useful, but it cannot substitute for a plausible prior or a check of its implications.

The posterior mean as a weighted average

For $p \sim \text{Beta}(a, b)$ and $Z \mid p \sim \text{Binomial}(n, p)$,

$$\mathbb{E}(p \mid z) = \frac{a+z}{a+b+n} = \frac{a+b}{a+b+n}\frac{a}{a+b} + \frac{n}{a+b+n}\frac{z}{n}.$$

- Prior mean: $a/(a+b)$; observed proportion: z/n.
- The weights reflect prior concentration $a+b$ and sample size n.
- For $a = b = 1$ and $z = 89, n = 108$, the weights are $2/110$ and $108/110$.

This explains why the posterior mean is close to, but not exactly, $89/108$.

When does the likelihood outweigh the prior?

Keep a Beta$(10, 10)$ prior, with mean 0.5 and concentration 20. Compare two samples with an observed admission rate of about 82.4%:

Sample	Prior weight $20/(20+n)$	Posterior	Posterior mean
Berkeley: 89/108	15.6%	Beta$(99, 29)$	0.773
Hypothetical: 890/1080	1.8%	Beta$(900, 200)$	0.818

- The same fixed prior has less influence as n grows; the posterior mean approaches the observed proportion.
- Thus a conjugate prior may be chosen partly for computational convenience, especially when large samples are available.

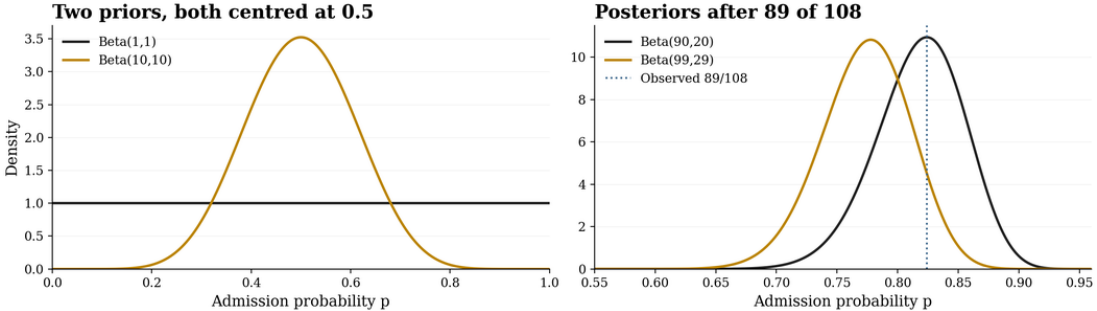
Summarise the worked posterior

For $p \mid z \sim \text{Beta}(90, 20)$:

- observed proportion: $89/108 = 0.824$;
- posterior mean: $90/110 = 0.818$;
- 95% equal-tail credible interval: $[0.741, 0.884]$;
- posterior probability: $\mathbb{P}(p > 0.80 \mid z) = 0.703$.

Prior-posterior sensitivity

Both priors have mean 0.5; they express different degrees of prior certainty.



- Beta$(1, 1) \to$ Beta$(90, 20)$: posterior mean 0.818.
- Beta$(10, 10) \to$ Beta$(99, 29)$: posterior mean 0.773.

The stronger prior pulls the posterior further toward 0.5 despite the same data.

From estimation to prediction

For m new applicants in the same setting, the predicted value of Z_{new} is given by its expected value, given the observed z average over uncertainty in p:

$$Z_{\text{new}} = \mathbb{E}\left[Z \mid z\right] = \sum_{j=0}^{m} j \int \underbrace{\binom{m}{j} p^j \left(1-p\right)^{m-j}}_{\text{p.m.f. Binomial}(m,p)} \underbrace{\pi\left(p \mid z\right)}_{\text{Post. dist. of } p} \, \mathrm{d}p$$

For $m = 10$ and $p \mid \boldsymbol{y} \sim \text{Beta}(90, 20)$:

$$\mathbb{E}(Z_{\text{new}} \mid z) = 10(0.818) = 8.18, \qquad 95\% \text{ predictive interval: } [5, 10].$$

The distribution of future counts includes both uncertainty about p and Binomial variation in new outcomes.

R: exact posterior summary and prediction

```r
# Summarise Beta(90, 20) without MCMC.
qbeta(c(0.025, 0.5, 0.975), 90, 20)
1 - pbeta(0.8, 90, 20)  # Posterior Pr(p > 0.8 | y).

# Exact predictive quantiles for 10 new applicants.
pred_pmf <- sapply(0:10, function(k)
  choose(10, k) * beta(90 + k, 20 + 10 - k) / beta(90, 20))
cdf <- cumsum(pred_pmf)
c(lower = which(cdf >= 0.025)[1] - 1,
  upper = which(cdf >= 0.975)[1] - 1)
```

Output (rounded): posterior quantiles $\mathbf{0.741}$, $\mathbf{0.820}$, $\mathbf{0.884}$; $\mathbb{P}(p > 0.8 \mid \boldsymbol{y}) = \mathbf{0.703}$; predictive interval $[\mathbf{5}, \mathbf{10}]$.

A model for the admission to Berkeley uni

- The Beta–Binomial analysis focused on the females applying to Department A;
- Suppose now we want to analyse the admission pattern to Berkeley uni by sex and department. Sex and department yield a 12-fold combination.

Returning to the grouped data

	Male applicants		Female applicants	
Department code	Admitted	Total	Admitted	Total
A	512	825	89	108
B	353	560	17	25
C	120	325	202	593
D	138	417	131	375
E	53	191	94	393
F	22	373	24	341
All departments	1,198	2,691	557	1,835

Overall admission proportions are 44.5% for men and 30.4% for women. The applicant mix differs by gender. The A–F labels are codes in the R dataset; the academic fields are not identified here.

Overall and department-specific admission rates

Department	Men admitted (%)	Women admitted (%)
A	62.1	82.4
B	63.0	68.0
C	36.9	34.1
D	33.1	34.9
E	27.7	23.9
F	5.9	7.0
All six departments combined	44.5	30.4

- Each A–F percentage is admitted applicants divided by all applicants of that gender *in that department*.
- The combined percentages *pool across the six departments*: 1,198/2,691 = 44.5% for men and 557/1,835 = 30.4% for women.
- Women have a higher rate in four of six departments, but a lower combined rate; the mix of departments applied to differs.

Previously: One model for each combination of sex and department

Six departments and two genders give 12 groups. For each applicant i in group k, let $y_{i,k}$ be the indicator variable equal to 1 in case of successful application and 0 otherwise, with success probability and p_k:

$$Y_{i,k} \mid p_k \sim \text{Bernoulli}(p_k), \qquad k = 1, \ldots, 12.$$

With independent outcomes and identically distributed by sex and department, the likelihood function is then:

$$L(p_k \mid \boldsymbol{y}) = \prod_{i=1}^{n_k} p_k^{y_{i,k}} (1 - p_k)^{1 - y_{i,k}}, \qquad i = 1, \ldots, n_k; \qquad k = 1, \ldots, 12$$

Each p_k is then estimated from its own group (for example, B/women has only 25 applicants). These separate analyses do not allow for statistical information to be shared across groups.

A Berkeley admission regression model

For applicant i, let $Y_i = 1$ if admitted and 0 otherwise; let $F_i = 1$ for women and 0 for men, and let d_i be the department applied for:

$$\begin{aligned} Y_i \mid p_i &\sim \text{Bernoulli}(p_i), \\ \text{logit}(p_i) &= \alpha + \beta F_i + \gamma_{d_i}, \qquad \gamma_A = 0. \end{aligned}$$

- α is the log odds for men in A; γ_d shifts the log odds for department d relative to A.
- β is the regression coefficient for the effect of sex on the logit rate of admission, all else being equal.

Priors, rates, and odds ratios

The following illustrative prior distributions are assumed

$$\alpha \sim N(0, 2.5^2), \qquad \beta \sim N(0, 2.5^2), \qquad \gamma_d \sim N(0, 2.5^2),\ d = B, C, D, E, F.$$

The female–male odds ratio is

$$\mathrm{OR}_F = \frac{\exp(\alpha + \beta)}{\exp(\alpha)} = \exp(\beta).$$

- $\mathrm{OR}_F < 1$: lower admission odds for women under this model.
- $\mathrm{OR}_F > 1$: higher admission odds for women under this model.

Before fitting, check the admission probabilities and rates implied by the priors.

Predicted values under the prior

The inverse logit converts log odds w into a probability:

$$p = \text{logit}^{-1}(w) = \frac{1}{1 + e^{-w}}.$$

Under the independent $N(0, 2.5^2)$ priors, consider

$$p_{A,M} = \text{logit}^{-1}(\alpha), \qquad p_{C,F} = \text{logit}^{-1}(\alpha + \beta + \gamma_C).$$

Group	Prior SD of log odds	Central 95% prior range for p
Department A / male	2.5	[0.007, 0.993]
Department C / female	$2.5\sqrt{3}$	[0.0002, 0.9998]

The central 95% range is between the 2.5th and 97.5th *prior* percentiles of p. For normal log odds with SD s, its endpoints are $[\text{logit}^{-1}(-1.96s), \text{logit}^{-1}(1.96s)]$.

The posterior distribution

The model has parameters $\theta = (\alpha, \beta, \gamma_B, \ldots, \gamma_F)$. The likelihood *kernel*[1] of the parameter given the observations is

$$L(\alpha, \beta, \gamma_B, \ldots, \gamma_F \mid \boldsymbol{y}) = \prod_{i=1}^{n} p_i(\theta)^{y_i} \{1 - p_i(\theta)\}^{1-y_i}.$$

Independent priors then give the joint posterior:

$$\begin{aligned}
\pi(\theta \mid y) &= \frac{L(\theta \mid \boldsymbol{y})\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d)}{\underbrace{\int L(\theta \mid \boldsymbol{y})\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d)\mathrm{d}\theta}_{f(\boldsymbol{y})}} \\
&\propto L(\theta \mid \boldsymbol{y})\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d).
\end{aligned}$$

[1] The omitted Binomial factors $\prod_k \binom{n_k}{y_k}$ and the normalising constant do not depend on θ; they change the overall scale, not the relative posterior density at different parameter values.

Marginalisation: averaging over other parameters

Suppose the research question concerns β. Its marginal posterior is

$$\pi(\beta \mid y) = \int \pi(\alpha, \beta, \boldsymbol{\gamma} \mid \boldsymbol{y})\, d\alpha\, d\boldsymbol{\gamma}.$$

For each candidate β, integrate over α and $\boldsymbol{\gamma}$, weighted by the joint posterior density.

- Uncertainty about department effects is retained in this integral; plugging in one estimate for each γ_d would discard it.
- We can similarly obtain a marginal for α, any γ_d, or a derived group probability p_k.

Prepare the grouped data in R

```r
# Convert the built-in three-way table to one row per gender and department.
counts <- as.data.frame(datasets::UCBAdmissions)

# Put admitted and rejected counts into separate columns.
wide <- reshape(counts, idvar = c("Gender", "Dept"),
                timevar = "Admit", direction = "wide")
names(wide) <- sub("^Freq\\.", "", names(wide))
names(wide)[names(wide) == "Admitted"] <- "admitted"
names(wide)[names(wide) == "Rejected"] <- "rejected"

# Create the binomial denominator and model covariates.
wide$n <- wide$admitted + wide$rejected
wide$female <- as.integer(wide$Gender == "Female")
wide$dept_id <- as.integer(factor(wide$Dept, levels = LETTERS[1:6]))
stopifnot(sum(wide$n) == 4526)
```

What can we report from the joint posterior?

The parameters of interest depend on the research question. The joint posterior supports inference on all seven coefficients and on functions of them.

- If all coefficients matter, report medians and 95% equal-tail intervals for $\alpha, \beta, \gamma_B, \ldots, \gamma_F$; state that A/men is the reference.
- For the gender comparison, report $\exp(\beta)$ and, if useful, $\mathbb{P}(\beta < 0 \mid y)$.
- For the effect of the department, report the value of the coefficients γ_d or the corresponding probabilities (by sex) with a posterior interval.

In every case, state the priors, the common-effect assumption and the limits of these grouped data.

Markov Chain Monte Carlo to approximate the posterior

- The Beta–Binomial posterior was available in closed form.
- For the logistic regression, the seven parameters $\theta = (\alpha, \beta, \gamma_B, \ldots, \gamma_F)$ have normal priors. The joint posterior has no standard closed form, and direct integration is complicated, sometimes not even feasible;
- Markov Chain Monte Carlo (MCMC) methods allow to draw *sequential* samples

$$\theta^{(1)}, \theta^{(2)}, \ldots, \theta^{(S)}$$

- Law of large numbers-based argument allow to prove that such samples are independent draws from the posterior distribution.
- In practice, the independence of samples need to be checked;
- In conclusion, we estimate $\pi(\theta \mid \boldsymbol{y})$ by drawing samples from it.

MCMC for approximations

In many models we can evaluate $p(y \mid \theta)\pi(\theta)$, but cannot normalise the posterior or integrate it analytically. MCMC constructs a *Markov chain* $\theta^{(1)}, \ldots, \theta^{(S)}$ whose target distribution is $\pi(\theta \mid y)$.

$$\underbrace{\frac{1}{S}\sum_{s=1}^{S} h(\theta^{(s)})}_{\text{average of draws}} \quad \approx \quad \underbrace{\int h(\theta)\pi(\theta \mid y)\, d\theta}_{\text{posterior expectation}}.$$

For the Berkeley example, choose $h(\theta) = \mathbf{1}\{\beta < 0\}$ to estimate $\mathbb{P}(\beta < 0 \mid y)$:

$$\mathbb{P}(\beta < 0 \mid y) = \int \mathbf{1}\{\beta < 0\}\, \pi(\theta \mid y)\, d\theta.$$

Hence, using MCMC draws $\theta^{(1)}, \ldots, \theta^{(S)}$, then

$$\widehat{\mathbb{P}}(\beta < 0 \mid y) = \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\{\beta^{(s)} < 0\}.$$

Random sequential draws with MCMC

Start with a parameter value $\theta^{(0)}$. A sampling rule uses its current value to produce $\theta^{(1)}$; then $\theta^{(1)}$ is used to produce $\theta^{(2)}$, and so on:

$$\theta^{(0)} \longrightarrow \theta^{(1)} \longrightarrow \theta^{(2)} \longrightarrow \cdots$$

For instance, a proposed move from an admission probability of 0.78 might be accepted as 0.82 or rejected, leaving the next state at 0.78.

- Each draw depends on the preceding state; the sequence is *not* a set of independent draws.
- Design the rule so its long-run distribution is the posterior; discard initial warm-up draws and check that multiple chains explore it well.

Metropolis–Hastings: propose, then accept or retain

Let $\tilde{\pi}(\theta \mid \boldsymbol{y}) = f(\boldsymbol{y} \mid \theta)\pi(\theta)$ be the unnormalised posterior. At iteration s:

1. Draw θ^* from a proposal distribution $q\left(\theta^* \mid \theta^{(s)}\right)$, $\theta^\star \sim q(\theta^\star \mid \theta^{(s)})$.
2. Accept θ^* with probability

$$a = \min\left\{1, \frac{\tilde{\pi}(\theta^\star \mid \boldsymbol{y}) q(\theta^{(s)} \mid \theta^\star)}{\tilde{\pi}(\theta^{(s)} \mid \boldsymbol{y}) q(\theta^\star \mid \theta^{(s)})}\right\}.$$

3. Otherwise retain $\theta^{(s)}$; a rejected move still counts as an iteration.

The unknown normalising constant cancels.

The posterior distribution for the Berkeley regression model

Let $\theta = (\alpha, \beta, \gamma_B, \ldots, \gamma_F)$. For the grouped logistic model, the log posterior kernel is

$$\ell(\theta) = \log \pi(\theta \mid \boldsymbol{y}) = \underbrace{\sum_{i=1}^{n} \left[y_i \log p_i(\theta) + (1 - y_i) \log\left(1 - p_i(\theta)\right) \right]}_{\text{Bernoulli log. likelihood}} \underbrace{- \frac{1}{2(2.5)^2} \sum_{j=1}^{7} \theta_j^2}_{\text{Normal log-lik. of prior}} + C.$$

At each iteration, propose a *joint* move in all seven coefficients, compare $\ell(\theta^\star)$ with $\ell(\theta)$, and accept or retain the current value. The normalising constant C cancels.

- Run four chains from different starting values, discard warm-up and inspect their traces before interpreting summaries.

R: target density and proposal for Berkeley

```r
# One row per observed department/gender cell, seven coefficient columns.
design <- model.matrix(~ female + Dept, data = wide)
n_par <- ncol(design)

# Log likelihood plus independent N(0, 2.5^2) log priors.
log_post <- function(theta) {
  eta <- drop(design %*% theta)
  sum(dbinom(wide$admitted, wide$n, plogis(eta), log = TRUE)) +
    sum(dnorm(theta, mean = 0, sd = 2.5, log = TRUE))
}

# The classical fit tunes proposals; it is not the Bayesian estimate.
fit_glm <- glm(cbind(admitted, rejected) ~ female + Dept,
               family = binomial(), data = wide)
step_U <- chol(vcov(fit_glm)) * 2.38 / sqrt(n_par)
```

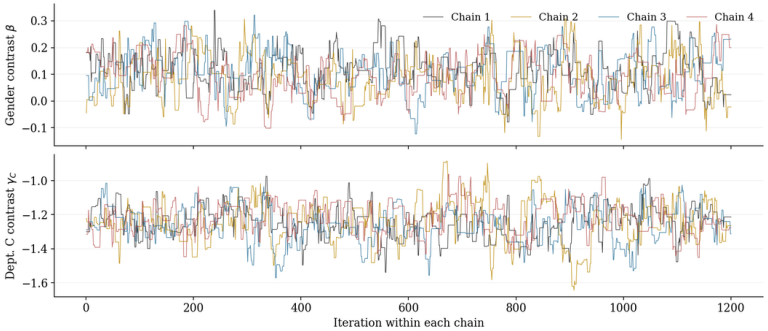
The row vector `rnorm(n_par) %*% step_U` in the next slide has covariance proportional to `vcov(fit_glm)`. This affects sampling efficiency, while the acceptance rule preserves the target posterior.

R: four Metropolis chains for the joint posterior

```
mh_one <- function(start, iterations = 18000, warmup = 3000) {
  theta <- start; value <- log_post(theta); accepted <- 0
  out <- matrix(NA_real_, iterations, n_par)
  colnames(out) <- colnames(design)
  for (s in seq_len(iterations)) {
    # Propose all coefficients jointly from a symmetric normal random walk.
    proposal <- theta + drop(rnorm(n_par) %*% step_U)
    trial <- log_post(proposal)
    if (log(runif(1)) < min(0, trial - value)) {
      theta <- proposal; value <- trial; accepted <- accepted + 1
    }
    out[s, ] <- theta  # Rejection repeats the previous state.
  }
  list(draws = out[(warmup + 1):iterations, , drop = FALSE],
       acceptance = accepted / iterations)
}
set.seed(1381)
mh <- lapply(seq_len(4), function(j) mh_one(rnorm(n_par, 0, .2)))
vapply(mh, function(z) z$acceptance, numeric(1))
```

Each retained row is one joint coefficient draw. The chain draws depend on their preceding state; mixing needs to be checked.

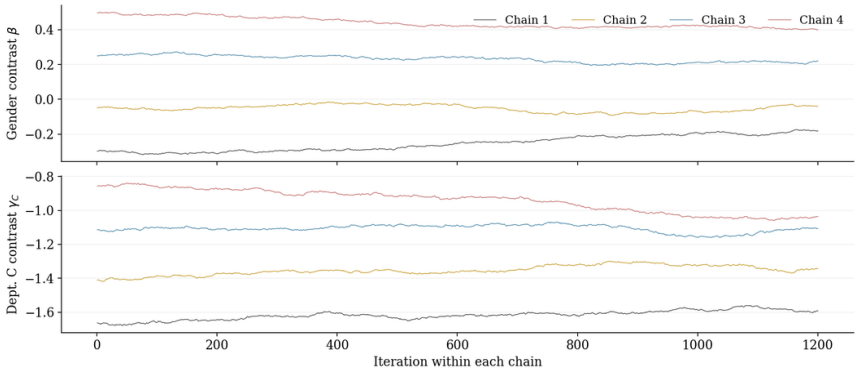
Bespoke Metropolis: inspect four chains



- The traces for β and γ_C should overlap, fluctuate within a stable range and show no persistent drift.
- Also check $\widehat{R}$ and effective sample size for *every* parameter; overlap in one short trace segment is not a proof of convergence.

Illustrative four-chain, hand-coded Metropolis run for the stated Berkeley posterior. Each panel shows 1,200 retained iterations; the JAGS chains have different random draws.

Counterexample: chains that do not mix



- The *same posterior* is targeted, but starts are dispersed, the proposal steps are very small, and only 1,200 iterations are run without warm-up.
- The chains stay separated: a pooled average or credible interval from this run is unreliable. High acceptance alone does not establish good sampling.

Additional checks on the MCMC sampler

- Inspect traces for *all seven* coefficients. Look for separation, slow drift, or long flat stretches from rejected moves.
- Compare the distributions from the beginning and end of each retained chain. They should tell a similar story.
- Examine autocorrelation: when a chain moves slowly, adjacent draws carry nearly the same information.
- If these checks fail, improve the proposal and run longer chains from dispersed initial values; then repeat the checks before reporting posterior summaries.

Two numerical diagnostics: $\widehat{R}$ and ESS

- $\widehat{R}$ compares variation *between* several chains with variation *within* them. Values close to 1 support agreement across chains; substantially larger values signal that they have not explored the same distribution.
- **Effective sample size (ESS)** accounts for dependence between successive draws. It is the approximate number of independent draws with comparable precision for estimating a posterior mean.
- With four chains of 15,000 retained draws, we store 60,000 values per coefficient, but the ESS can be much smaller if successive draws are strongly correlated.
- Examine both diagnostics for every coefficient alongside traces. Neither number alone proves convergence or model fit.

In this lecture, `coda::gelman.diag()` reports a classical potential scale-reduction estimate of $\widehat{R}$; Stan later provides modern split and rank-based diagnostics.

R: diagnose the well-mixed and poor runs

```r
# Convert four independently run R chains to a coda mcmc.list.
good_coda <- do.call(coda::mcmc.list,
  lapply(mh, function(z) coda::mcmc(z$draws)))

# Compare between-chain with within-chain variation for every column.
coda::gelman.diag(good_coda, autoburnin = FALSE)$psrf

# Estimate the independent-draw equivalent for every coefficient.
coda::effectiveSize(good_coda)

# Repeat for deliberately inadequate 1,200-iteration runs.
poor_coda <- do.call(coda::mcmc.list,
  lapply(mh_poor, function(z) coda::mcmc(z$draws)))
coda::gelman.diag(poor_coda, autoburnin = FALSE)$psrf
coda::effectiveSize(poor_coda)
```

The full script creates `mh_poor` by using `step_scale = .02` and dispersed starting values. Do not use those draws for inference.

Summary of the posterior inference

The parameters of interest depend on the research question. The joint posterior supports inference on all seven coefficients and on functions of them.

- If all coefficients matter, report medians and 95% equal-tail intervals for $\alpha, \beta, \gamma_B, \ldots, \gamma_F$; state that A/men is the reference.
- For the gender comparison, report $\exp(\beta)$ and, if useful, $\mathbb{P}(\beta < 0 \mid y)$. Exponentiate equal-tail interval endpoints to give an odds-ratio interval.
- For a group k, report the admission probability $p_k = \text{logit}^{-1}(\alpha + \beta F_k + \gamma_{d[k]})$ with a posterior interval.

In every case, state the priors, the common-effect assumption and the limits of these grouped data.

JAGS: Just Another Gibbs Sampler

- **JAGS** (Just Another Gibbs Sampler) reads a probabilistic model written in BUGS notation: priors, likelihood and named derived quantities.
- In R, `rjags` passes the model and data to JAGS. JAGS runs sequential MCMC updates, with an update method selected for each part of the model.
- The output is several chains of posterior draws. `coda` provides summaries, trace plots and convergence diagnostics.
- From the draws we estimate intervals, probabilities and predictions, including replicated Berkeley tables.

For the logistic Berkeley model, not every JAGS update is necessarily a direct Gibbs draw. Later we use a normal model to show when full conditionals make Gibbs updates simple.

See the JAGS user manual and the `rjags` reference manual.

The complete JAGS workflow: Berkeley admissions

1. **Define the question:** estimate admission probabilities and the within-department gender association from the 12 grouped cells.
2. **Specify the model:** Binomial counts, logit probabilities, seven unknown coefficients and their normal priors.
3. **Compute:** start four JAGS chains, adapt and warm up, then retain posterior draws.
4. **Diagnose:** inspect traces of every coefficient, $\widehat{R}$ and effective sample size.
5. **Infer and check:** summarise marginal posteriors; simulate another 12-cell table and compare it with the observations.
6. **Predict:** use each posterior draw to predict offers among ten future applicants with a specified profile.

JAGS model for the Berkeley table

```
# JAGS uses precision as the second argument to dnorm.
jags_text <- "
model {
  alpha ~ dnorm(0, prior_precision)
  beta_female ~ dnorm(0, prior_precision)
  beta_dept[1] <- 0
  for (d in 2:6) { beta_dept[d] ~ dnorm(0, prior_precision) }
  for (k in 1:K) {
    admitted[k] ~ dbin(p[k], applicants[k])
    logit(p[k]) <- alpha + beta_female * female[k] + beta_dept[dept[k]]
  }
}
"
```

The zero reference effect identifies Department A. The remaining department coefficients are relative shifts on the log-odds scale.

R: describe the Berkeley data to JAGS

```
# The names below match those in the JAGS model on the previous slide.
jags_data <- list(
  K = nrow(wide),                    # 12 department/gender cells
  admitted = as.integer(wide$admitted), # offers in each cell
  applicants = as.integer(wide$n),      # applications in each cell
  female = as.integer(wide$female),     # 0 = man; 1 = woman
  dept = as.integer(wide$dept_id),      # A = 1, ..., F = 6
  prior_precision = 1 / 2.5^2          # inverse prior variance
)
```

`as.integer()` passes whole-number counts and category codes. The prior $N(0, 2.5^2)$ becomes precision $1/2.5^2$ in JAGS.

R: start four JAGS chains

```r
# seq_len(4) labels the chains 1, 2, 3 and 4.
# Give each chain a distinct, reproducible random-number seed.
inits <- lapply(seq_len(4), function(j)
  list(.RNG.name = "base::Wichmann-Hill", .RNG.seed = 20000L + j))

# Supply the JAGS model text held in jags_text.
tc <- textConnection(jags_text)

# Compile the model, attach data and adapt four chains.
jm <- rjags::jags.model(tc, data = jags_data, inits = inits,
                        n.chains = 4, n.adapt = 1000)
close(tc) # Close the temporary connection to the model text.
```

`n.adapt` lets JAGS tune its samplers for 1,000 iterations. The next slide runs additional warm-up iterations before saving draws.

R: retain four posterior chains

```r
# Let the chains move before storing the draws used for inference.
update(jm, n.iter = 1000)

# Monitor exactly seven unknown coefficients; A is the zero reference.
jags_draws <- rjags::coda.samples(jm,
  variable.names = c("alpha", "beta_female", "beta_dept[2:6]"),
  n.iter = 4000)

# Department A has coefficient zero by definition, not by estimation.
coef_names <- c("alpha", "beta_female",
                paste0("beta_dept[", 2:6, "]"))
coef_chains <- jags_draws[, coef_names]
summary(coef_chains)  # Posterior mean, SD, quantiles by coefficient.
```

Four chains $\times$ 4,000 retained iterations give 16,000 stored draws per coefficient. Diagnose the chains before combining their draws.

R: inspect all seven JAGS traces

```r
# Take exactly the seven estimated coefficients from the JAGS output.
coef_names <- c("alpha", "beta_female",
                paste0("beta_dept[", 2:6, "]"))
coef_chains <- jags_draws[, coef_names]

# Plot all seven traces in R: a separate panel per coefficient.
# In each panel the four colours represent the four chains.
chain_cols <- c("black", "goldenrod3", "steelblue", "firebrick3")
coda::traceplot(coef_chains, col = chain_cols)

# Also check between-chain agreement and within-chain dependence.
print(coda::gelman.diag(coef_chains, autoburnin = FALSE)$psrf)
print(coda::effectiveSize(coef_chains))
```

Paste this code into an R session after running the preceding slides. The trace plots appear in R's graphics window; no image file is required.

What to inspect in the JAGS output

1. `summary(draws)` gives a mean, SD and quantiles for each monitored variable. For `OR_female`, the 2.5% and 97.5% quantiles form an equal-tail credible interval.
2. `plot(draws)` shows trace and density plots. Check that the four traces mix and show no persistent drift.
3. Apply `gelman.diag(draws[,coef_names])` and `effectiveSize(draws[,coef_names])`. Values of $\widehat{R}$ close to 1 support agreement across chains; effective sample size accounts for autocorrelation.

Output is conditional on the stated model and priors. A well-mixed chain alone does not imply that the logistic model fits the admission counts.

R: summarise every coefficient from JAGS draws

```r
# Combine the retained draws from all four chains, after diagnostics.
mat <- do.call(rbind, lapply(jags_draws, as.matrix))
coef_names <- c("alpha", "beta_female",
                paste0("beta_dept[", 2:6, "]"))

# Compute a median, SD and equal-tail 95% interval per coefficient.
post_summary <- t(vapply(coef_names, function(name) {
  x <- mat[, name]
  c(median = median(x), SD = sd(x),
    low = unname(quantile(x, .025)),
    high = unname(quantile(x, .975)))
}, numeric(4)))
print(round(post_summary, 2))

# The table reports only the seven estimated coefficients.
# Cell probabilities are calculated later for checking and prediction.
```

Posterior summary: all seven coefficients

Parameter	Median	SD	95% equal-tail credible interval
α	0.57	0.07	$[0.44,\ 0.71]$
β	0.09	0.08	$[-0.07,\ 0.25]$
γ_B	-0.03	0.11	$[-0.24,\ 0.19]$
γ_C	-1.25	0.11	$[-1.45,\ -1.04]$
γ_D	-1.28	0.10	$[-1.49,\ -1.08]$
γ_E	-1.72	0.13	$[-1.97,\ -1.48]$
γ_F	-3.29	0.16	$[-3.62,\ -2.98]$

α describes men in A; β is the common gender contrast; each γ_d compares department d with A at the same gender.

Rounded illustrative four-chain MCMC summaries for the stated $N(0, 2.5^2)$ priors. Repeated runs and JAGS sampling may change the last digit.

Interpret the posterior coefficients

The probability of admission for a man applying to Department A is $\text{logit}^{-1}(\alpha)$; α is a baseline *log-odds*.

- β captures the effect of sex in the log-odds admission rate. Draws of $\exp(\beta)$ give a median odds ratio of about 1.10 and a 95% interval $[0.94, 1.29]$.
- γ_C is the Department C–A log-odds difference at the same gender. A value near -1.25 corresponds to odds multiplied by about $\exp(-1.25) = 0.29$.
- The credible intervals describe coefficient uncertainty under a common gender effect.

Transform *each posterior draw* to a probability or odds ratio before calculating an interval for that derived quantity.

Read a posterior summary: Berkeley

Quantity	Approximate summary	Interpretation
β	median 0.09; 95% CrI $[-0.07, 0.25]$	adjusted log-odds contrast
$\exp(\beta)$	median 1.10; 95% CrI $[0.94, 1.29]$	adjusted odds ratio, women versus men
$\mathbb{P}(\beta < 0 \mid y)$	0.13	probability of lower adjusted odds

The interval crosses 1 on the odds-ratio scale. The posterior does not establish equal odds; it allows modest differences in either direction.

Rounded summaries from an illustrative MCMC run for the stated model and priors; sampling and diagnostic steps were shown earlier.

Sensitivity to the prior distribution

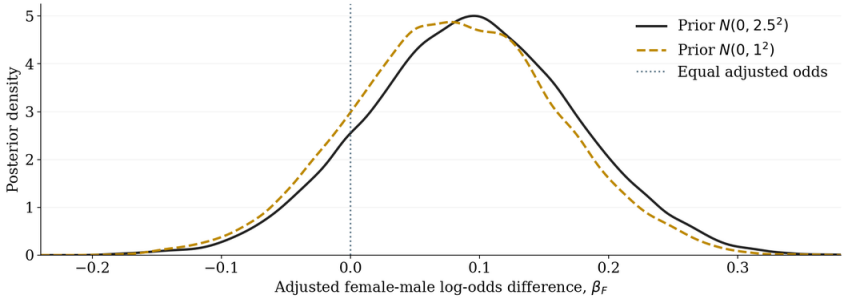
Does the adjusted sex comparison change if each coefficient receives a narrower $N(0, s^2)$ prior? Keep the same likelihood and compare:

Prior SD s	Median β	Adjusted OR (95% interval)	$\mathbb{P}(\beta < 0 \mid y)$
2.5	0.09	1.10 $[0.94, 1.29]$	0.13
1	0.08	1.08 $[0.93, 1.27]$	0.16

For this one comparison the inference is similar, though the probability of a negative contrast changes. Sensitivity of one coefficient does not validate the common-effect model; check its predicted cell counts as well.

A graphical prior-sensitivity check

Compare sampled posteriors for the adjusted log-odds contrast β, keeping the likelihood fixed and changing the prior scale.



- Compare each curve's centre, spread and area to the left of zero.
- The inference changes little; checking the *model's fit* requires replicating the observed table.

Density estimates from illustrative MCMC draws under the same model as the preceding table.

Posterior predictive checks for the Berkeley table

With the chains checked, ask whether the fitted model could produce data like the observed 12-cell table. For every retained draw s and cell k,

$$p_i^{(s)} = \text{logit}^{-1}(\alpha^{(s)} + \beta^{(s)} F_k + \gamma_{d[k]}^{(s)}),$$
$$Y_i^{\text{rep},(s)} \sim \text{Bernoulli}(p_k^{(s)}).$$

- A number of draws equal to the total number of applicant for each sex and department yields a new admissions table under the fitted model.
- Compare the observed z_k with the 2.5th and 97.5th percentiles of its replicated counts. Look for systematic differences by department or gender.

An observed discrepancy: Department A

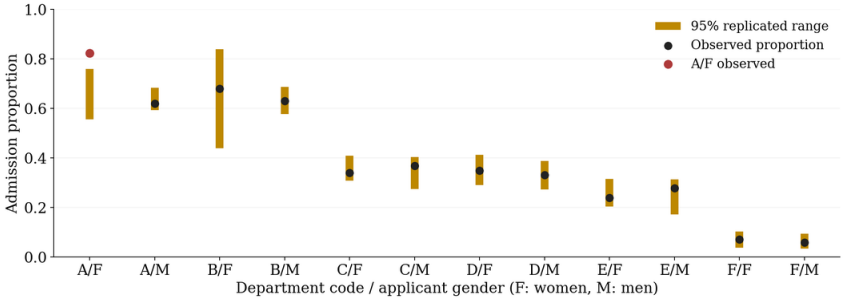
For the common gender-effect model fitted earlier, compare observed offers with the distribution of replicated offers at the *same* cell sizes.

Department / gender	Applicants	Observed offers	95% predictive interval
A / male	825	512	[490,564]
A / female	108	89	[60,82]
E / female	393	94	[80,124]

The observed 89 offers for women in A exceed the upper predictive limit. One outlying cell among 12 could occur by chance, but it prompts a closer look: does a gender-by-department interaction improve the fit?

Graphical model check: observed versus replicated

Compare each observed admission proportion with replicated counts divided by the same group size.



- The A/women observed proportion lies above its 95% predictive interval.
- Investigate a varying gender effect; isolated deviations among 12 checks are possible.

In-sample replicated counts under the common-effect model, using MCMC draws and Binomial simulations.

R: reproduce the Berkeley check

```r
# Use the JAGS chains fitted and checked in the earlier section.
# Extract retained cell probabilities from its four MCMC chains.
mat <- do.call(rbind, lapply(jags_draws, as.matrix))
p_names <- paste0("p[", seq_len(nrow(wide)), "]")
p_ppc <- mat[, p_names, drop = FALSE]
set.seed(20261026)  # Reproduce the Binomial simulations.

# Replicate counts with the original applicant numbers.
y_ppc <- vapply(seq_len(nrow(wide)), function(k)
  rbinom(nrow(p_ppc), size = wide$n[k], prob = p_ppc[, k]),
  numeric(nrow(p_ppc)))

# Compare the observed counts with their predictive limits.
limits <- apply(y_ppc, 2, quantile, probs = c(0.025, 0.975))
print(data.frame(wide[c("Dept", "Gender", "admitted")],
                 low = limits[1, ], high = limits[2, ]))
```

Workflow: after checking the four chains, simulate a table for each draw and compare observed counts with the intervals.

Posterior predictive distribution

- For a new male applicant to department B, $Y_{\text{new,M,B}}$, the probability of admission is given by:

$$\begin{aligned}\Pr(Y_{\text{new,M,B}} = 1 \mid \boldsymbol{y}) &= \int \frac{1}{1 + e^{-(\alpha+\gamma_B)}} \pi(\theta \mid \boldsymbol{y})\, d\theta \\ &\approx \frac{1}{S} \sum_{s=1}^{S} \frac{1}{1 + e^{-\left(\alpha^{(s)}+\gamma_B^{(s)}\right)}}\end{aligned}$$

- Similarly, it is possible to derive the expected number of successful applications out of n_{new} applicants.

R: predict admissions for a new group

```r
# Find the observed cell that defines the new applicant profile.
k <- which(wide$Dept == "A" & wide$Gender == "Female")

# Use the JAGS probabilities for that profile, one per posterior draw.
mat <- do.call(rbind, lapply(draws, as.matrix))
p_new <- mat[, paste0("p[", k, "]")]

# Conditional on each draw, simulate offers among 10 new applicants.
set.seed(20261028)
y_new <- rbinom(length(p_new), size = 10, prob = p_new)

# Report mean and central 95% range of the predictive distribution.
mean(y_new)
quantile(y_new, c(.025, .5, .975))
```

These predictions apply to a future group with the same recorded gender and department under an unchanged admissions process. A new cohort would be needed to assess out-of-sample accuracy.

Illustrative output: mean ≈ 6.6; 95% predictive interval $[4, 9]$.

The predictive answer depends on the fitted model

For ten future women applying to Department A, the two analyses give different answers because they make different assumptions:

Model	Predictive mean	95% predictive interval
Department A women alone: Beta–Binomial	8.18	$[5, 10]$
All 12 cells: common-effect logistic model	≈ 6.6	$\approx [4, 9]$

The one-cell model estimates this group's rate from its 89 offers among 108 applicants. The joint model imposes one gender coefficient across departments; its replicated A/women counts tend to fall below the observed 89.

Before using the joint model to forecast a new cohort, investigate that lack of fit and consider a department-specific gender effect.

Model fit and predictive ability are different

- **Model check:** simulate another 12-cell table with the same cell sizes, then compare counts, department rates and gender patterns with the observed table.
- **Predictive assessment:** use a later admissions cohort, held out before fitting, and evaluate calibration or predictive log scores on those new data.
- A model can match the historical table yet predict a different future cohort poorly.

Gibbs sampling: The normal distribution with unknown mean and variance

- First of all, the Gibbs sampler is a computational method to draw random samples of multidimensional parameters from their joint posterior distribution;
- Again, such posterior distribution cannot be obtained in closed form;
- However, in some cases, for some parameters there exist a closed form conditional posterior distribution.
  - Therefore, each parameter (or block of parameters) is sequentially drawn from the (conditional) posterior distribution;

Example: distribution of income among blue collar employees

For example, suppose that the log-income Y follows a Normal distribution with mean μ and variance σ^2, both unknown. Given a sample of $n = 40$ blue collar employees, the Bayesian analysis of these data is as follows:

- Probability model for Y_i,

$$Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$$

- Prior distributions for $\left(\mu, \sigma^2\right)$:

$$\mu \sim N(m_0, s_0^2), \quad \sigma^2 \sim \text{InverseGamma}(a_0, b_0),$$

- The *joint* posterior of (μ, σ^2) is given by:

$$\pi\left(\mu, \sigma^2 \mid \boldsymbol{y}\right) \propto \underbrace{\pi\left(\mu\right)}_{N\left(m_0, s_0^2\right)} \underbrace{\pi\left(\sigma^2\right)}_{IG(a_0, b_0)} \underbrace{\prod_{i=1}^{n} f\left(y_i \mid \mu, \sigma^2\right)}_{\text{Likelihood}}$$

- A closed form for $\pi\left(\mu, \sigma^2 \mid \boldsymbol{y}\right)$ cannot be obtained.

Example: distribution of income among blue collar employees (2)

However:

- $\pi\left(\mu \mid \sigma^2, \boldsymbol{y}\right)$, is the density of a Normal distribution with variance:

$$V_\mu = \left(\frac{1}{s_0^2} + \frac{n}{\sigma^2}\right)^{-1}.$$

and mean:

$$M_\mu = V_\mu \left(\frac{m_0}{s_0^2} + \frac{n\bar{y}}{\sigma^2}\right).$$

- $\pi\left(\sigma^2 \mid \mu, \boldsymbol{y}\right)$ is another InverseGamma distribution with shape

$$a_0 + \frac{n}{2}$$

and scale:

$$b_0 + \frac{1}{2}\sum_i (y_i - \mu)^2$$

The two full conditional distributions

We thus obtain that μ and σ^2 are *conditionally conjugate* (when data are normally distributed). The inverse-gamma with shape a and scale b for a random variable X (in our case given by σ^2) has density proportional to

$$\pi(x) \propto x^{-a-1} \exp(-b/x) \quad \text{for} \quad x > 0$$

The conditional conjugacy is computationally convenient. Indeed, to compute $\pi\left(\mu, \sigma^2 \mid \boldsymbol{y}\right)$, start at $(\mu^{(0)}, \sigma^{2(0)})$. Then, at iteration s:

$$\mu^{(s)} \sim \pi(\mu \mid \sigma^{2(s-1)}, y),$$
$$\sigma^{2(s)} \sim \pi(\sigma^2 \mid \mu^{(s)}, y).$$

- Use the old variance to draw a new mean; immediately use the new mean to draw a new variance.
- Repeat, retaining pairs after warm-up.
- JAGS can automate this process.

R: two Gibbs updates

```r
# Simulate 40 log annual incomes (euros) for the Gibbs example.
set.seed(41)
y <- rnorm(40, mean = 10, sd = 0.4)
n <- length(y)

# Set prior values and initialise the chain.
m0 <- 10; s0_sq <- 1; a0 <- 2; b0 <- 0.16
mu <- mean(y); sigma2 <- var(y)
draws <- matrix(NA_real_, nrow = 12000, ncol = 2)

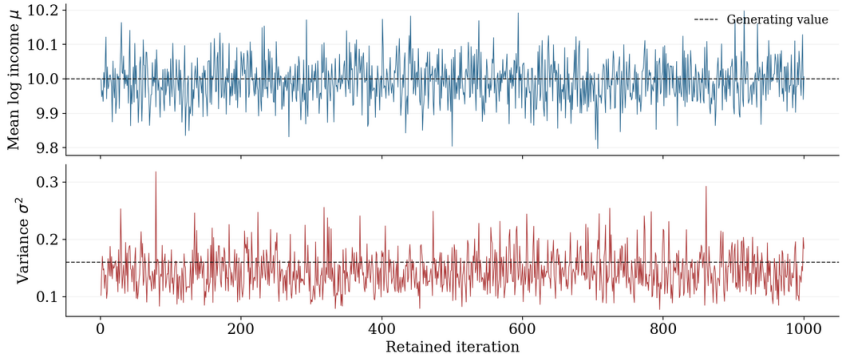
# Alternate draws from the full conditional distributions.
for (s in 1:nrow(draws)) {
  V_mu <- 1 / (1 / s0_sq + n / sigma2)
  M_mu <- V_mu * (m0 / s0_sq + sum(y) / sigma2)
  mu <- rnorm(1, mean = M_mu, sd = sqrt(V_mu))
  shape <- a0 + n / 2
  rate <- b0 + sum((y - mu)^2) / 2
  sigma2 <- 1 / rgamma(1, shape = shape, rate = rate)
  draws[s, ] <- c(mu, sigma2)
}
```

The two-parameter normal model in JAGS

```
# The same data y, n and priors as in the manual Gibbs example.
normal_model <- "model {
  for (i in 1:N) { y[i] ~ dnorm(mu, precision) }
  mu ~ dnorm(m0, 1 / s0_sq)
  precision ~ dgamma(a0, b0)
  sigma2 <- 1 / precision
  mean_income <- exp(mu + sigma2 / 2)
}"
tc <- textConnection(normal_model)
jm <- rjags::jags.model(tc, data = list(
  N = length(y), y = y, m0 = 10, s0_sq = 1,
  a0 = 2, b0 = 0.16), n.chains = 4, n.adapt = 1000)
close(tc)
update(jm, 1000)
draws <- rjags::coda.samples(jm, c("mu", "sigma2", "mean_income"),
                             n.iter = 4000)
summary(draws)     # Compare with the hand-coded Gibbs draws.
```

A gamma prior with shape a_0 and rate b_0 for precision $1/\sigma^2$ corresponds to the stated inverse-gamma prior for σ^2 (shape a_0, scale b_0).

Gibbs traces for log annual income



- The mean log income moves around 10; the variance is on the log-income scale.
- After warm-up, check stable exploration. A single chain cannot establish convergence; compare several chains in practice.

First 1,000 of 10,000 retained iterations in an independent illustrative Gibbs run.

Posterior summaries: log income and annual income

For each retained joint draw $(\mu^{(s)}, \sigma^{2(s)})$, the conditional mean annual income is $e^{\mu^{(s)}+\sigma^{2(s)}/2}$. Summarise those transformed draws rather than exponentiating a marginal interval for μ.

Parameter	Posterior median	Illustrative 95% interval
Mean log income μ	9.99	[9.87,10.11]
Variance σ^2	0.14	[0.09,0.22]
Mean income $e^{\mu+\sigma^2/2}$	EUR 23,400	[EUR 20,900, EUR 26,800]

The 95% intervals enclose the central 95% of the corresponding posterior draws. The income summary is in euros; μ and σ^2 refer to log euros.

Rounded values from a separate simulation of the stated model and priors; R and JAGS use different random draws.

More complex models

Suppose we had individual Berkeley application records: applicant i's admission $Y_i \in \{0, 1\}$, gender F_i, department d_i, prior grade x_i and school $s(i)$. An illustrative extension is

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \alpha + \beta_F F_i + \gamma_{d_i} + \delta x_i + u_{s(i)}, \quad u_s = \tau z_s,\ z_s \sim N(0, 1).$$

- x_i records a measured applicant feature. The latent u_s captures remaining differences among schools; $\tau > 0$ controls their spread. Set $\gamma_A = 0$.
- With many schools, the posterior has many related z_s as well as common coefficients. Gradient-based HMC can explore such a posterior effectively; the non-centred form can help its geometry.

Hamiltonian Monte Carlo: use the posterior gradient

- Sequential samples from a posterior distribution using MCMC methods, such as Metropolis-Hastings and Gibbs sampling (when possible) are likely to yield correlated samples in situations where the parameter space is high-dimensional;
- The Hamiltonian Monte Carlo method allows to (partially) overcome this issue by using a proposal distribution based on the gradient of the log-posterior distribution;
- This technique allows for a more efficient exploration of the parameter space and for a better mixing across chains with different starting points.

Hamiltonian Monte Carlo: use the posterior gradient

- In the hypothetical applicant-level model, let θ collect all unknown continuous parameters (including z_s and τ), and let $\boldsymbol{y}$ denote the observed admission decisions.
- HMC draws an artificial momentum $r \sim N(0, M)$ of the same dimension as θ:

$$H(\theta, r) = U(\theta) + K(r), \quad U(\theta) = -\log \pi(\theta \mid y), \quad K(r) = \tfrac{1}{2} r^{\mathsf{T}} M^{-1} r.$$

  where H is total energy, U is negative log posterior density (with π denoting that density), K is kinetic energy, and M is a positive-definite scale matrix. Again, the posterior normalising constant cancels from the calculations.

  - The posterior gradient gives a direction; numerical steps move θ and r together to propose a distant value of θ.
  - An accept/reject step corrects numerical approximation error. This can move more efficiently than an unguided random walk, particularly with many correlated parameters.

HMC needs differentiable posterior density in continuous parameters. See the Stan Reference Manual on HMC.

NUTS and Stan

Stan is a software with R interface RStan which allows to draw samples from a complex posterior distribution:

- Stan implements the No-U-Turn Sampler (NUTS), which is an adaptive form of HMC, requiring little or no user input.
- Stan translates a probability model into log density and gradients, then runs NUTS by default for MCMC sampling.
- Users still choose the model, priors, parameterisation and data checks.

Diagnostics to inspect: $\widehat{R}$, ESS, trace plots.

Stan from R: the practical workflow

- **Stan** is the modelling language and inference engine. Specify data, unknown continuous parameters, priors and likelihood in a `.stan` program.
- **RStan** and **CmdStanR** are two R interfaces. We use CmdStanR: it compiles the program through a separate CmdStan installation and returns draws to R.
- Build a named R list whose entries match the Stan `data` block. Compile with `cmdstan_model()`, sample four NUTS chains with `$sample()`, and inspect summaries and sampler warnings.
- Extract coefficient draws to calculate posterior odds ratios, intervals and predictions, as with JAGS.

See the official CmdStanR getting-started guide.

Set up CmdStanR for the practical example

```r
# Run once: install the R interface from the Stan repository.
install.packages("cmdstanr", repos = c(
  "https://stan-dev.r-universe.dev", getOption("repos")))

# A working C++ toolchain is needed to compile Stan models.
cmdstanr::check_cmdstan_toolchain()

# Install the separate CmdStan engine, then verify it is available.
cmdstanr::install_cmdstan(cores = 2)
cmdstanr::cmdstan_version()

# The companion script contains stan_code and stan_data.
# Run that script after setup to fit and diagnose the example.
```

RStan is another R interface to Stan; the following slides consistently use CmdStanR. See the official CmdStanR installation guide for your operating system.

Stan model: data and parameters

```
data {
  int<lower=1> K;
  array[K] int<lower=0> admitted;
  array[K] int<lower=0> applicants;
  array[K] int<lower=0, upper=1> female;
  array[K] int<lower=1, upper=6> dept;
}
parameters {
  real alpha;
  real beta_female;
  vector[5] beta_dept;
}
transformed parameters {
  vector[K] eta;
  for (k in 1:K) {
    eta[k] = alpha + beta_female * female[k];
    if (dept[k] > 1) eta[k] += beta_dept[dept[k] - 1];
  }
}
```

The integer arrays describe the 12 grouped cells. Department A is the reference; the five-element vector stores effects for Departments B–F.

Stan model: priors and likelihood

```
model {
  alpha ~ normal(0, 2.5);
  beta_female ~ normal(0, 2.5);
  beta_dept ~ normal(0, 2.5);
  admitted ~ binomial_logit(applicants, eta);
}
generated quantities { real OR_female = exp(beta_female); }
```

The likelihood is Binomial on the count scale. Stan uses the logit parameterisation for numerical stability; the generated quantity converts the gender coefficient to an odds ratio for every posterior draw.

Fit and diagnose with CmdStanR

```r
# Once in an R installation: install CmdStanR from the Stan repository,
# then check the C++ toolchain and run cmdstanr::install_cmdstan().
# The complete Stan program is stan_code in the companion R script.
stan_file <- tempfile(fileext = ".stan")
writeLines(stan_code, stan_file)    # Write the Stan language model.
stan_model <- cmdstanr::cmdstan_model(stan_file) # Compile it.

# Fit four NUTS chains with warm-up, then inspect diagnostics.
fit <- stan_model$sample(data = stan_data, seed = 20261026,
  chains = 4, parallel_chains = 4,
  iter_warmup = 1000, iter_sampling = 2000)
fit$summary(variables = c("beta_female", "OR_female")) # R-hat, ESS, intervals.
fit$diagnostic_summary()         # Divergences and other sampler warnings.
```

The R package `cmdstanr`, CmdStan and a working C++ toolchain are required.

Stan: turn draws into a conclusion

```r
# Flatten retained chains to a matrix of posterior draws.
mat <- posterior::as_draws_matrix(fit$draws(
  variables = c("alpha", "beta_female", "OR_female")))

# Report an equal-tail interval and the probability of a negative contrast.
quantile(mat[, "OR_female"], c(.025, .5, .975))
mean(mat[, "beta_female"] < 0)

# Probability for a new female applicant to Department A.
p_A <- plogis(mat[, "alpha"] + mat[, "beta_female"])

# For each posterior state, draw a count among ten similar applicants.
y_new <- rbinom(length(p_A), size = 10, prob = p_A)
quantile(y_new, c(.025, .5, .975))
```

Each prediction propagates coefficient uncertainty and outcome variation, as in the earlier posterior predictive formula.

Divergences are a model diagnostic

- A divergent transition means the numerical trajectory may not have followed the target distribution accurately in some region.
- Inspect where they occur; consider stronger prior information, rescaling, or a better parameterisation.
- Refit and recheck $\widehat{R}$, ESS, trace plots and Monte Carlo error.

For this low-dimensional example, a well-fitting model should usually sample without divergences; the diagnostics must still be checked.

Choosing a computational approach

Method	Main idea	Best teaching point
Conjugate update	Algebraic posterior in the same family	Exact Beta–Binomial analysis; no sampler needed
Metropolis–Hastings	Propose a move and accept or reject it	General and transparent; proposal tuning matters
Gibbs	Draw each parameter from its full conditional	Conditional conjugacy can help; mixing may be slow
HMC / NUTS	Gradient-guided proposals	Efficient for continuous models; diagnose divergences

Practical Bayesian workflow

1. **Question and model:** define the probability model for the data and the statistical research question;
2. **Before fitting:** simulate the quantities of interest based on random parameter draws from their prior distribution and revise implausible assumptions.
3. **Computation:** fit the posterior; check trace plots, $\widehat{R}$, effective sample size and Monte Carlo error.
4. **Interpretation of the results**;
5. **Revisit and report:** refit, check sensitivity to the prior distribution, especially in case of smaller samples and assess prediction on a held-out part of the data.

Recap and takeaways

- Bayes' rule combines a likelihood and prior to ground the statistical analysis on the posterior distribution.
- The posterior supports estimation, uncertainty statements and prediction.
- Conjugacy can give an exact posterior; MCMC approximates it when direct calculation is difficult.
- Metropolis–Hastings, Gibbs and HMC use different transition strategies;
- JAGS and Stan turn a model specification into posterior draws. The ultimate judgment of the outcome is still on the analyst;
- For the Berkeley table, adjustment for department changes the question from a pooled comparison to a model-based within-department association.

Further topics: computation with large datasets

MCMC replaces a difficult integral by averages of posterior draws, but obtaining those draws can be costly.

- Each iteration may evaluate a likelihood (and, for HMC, its gradient) using a large dataset.
- The chain progresses sequentially; warm-up, slow mixing and multiple chains add computation.
- With many observations and latent variables, the time needed for sufficiently accurate posterior summaries may be substantial.

This motivates *approximate inference by optimisation*, among other approaches. The useful comparison is computational cost *and* the accuracy needed for the scientific question.

Variational inference: approximate by optimisation

Choose a tractable family of distributions $q_\phi(\theta)$. Find the parameter value which ensures that $q_\phi(\theta)$ is as close as possible to the posterior distribution by minimising the

$$\mathrm{KL}\big(q_\phi(\theta) \,\|\, \pi(\theta \mid y)\big) = \mathbb{E}_{q_\phi}\left[\log \frac{q_\phi(\theta)}{\pi(\theta \mid y)}\right].$$

Because the posterior normalising constant does not depend on ϕ, this is equivalent to maximising the *evidence lower bound* (ELBO):

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(y, \theta)] - \mathbb{E}_{q_\phi}[\log q_\phi(\theta)].$$

- Speed can improve, but restrictive families may miss dependence across elements of θ or understate uncertainty. Check the approximation when it matters.

See Blei, Kucukelbir and McAuliffe (2017).

Example: topic models and variational inference

Suppose we have a large collection of documents. In *latent Dirichlet allocation* (LDA), each document mixes topics, and each topic assigns probabilities to words. The topic proportions and assignments are latent.

- **Inferential task:** estimate which topics occur in each document and which words characterise each topic.
- **Computational challenge:** jointly updating all these latent quantities across many documents is difficult.
- **Landmark:** Blei, Ng and Jordan (2003) used variational methods for approximate inference in LDA.
- **At larger scale:** stochastic variational inference uses subsets of documents per update; Hoffman et al. (2013) analysed corpora with millions of documents.

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Data and software documentation: `https://search.r-project.org/R/refmans/datasets/html/UCBAdmissions.html`
JAGS: `https://mcmc-jags.sourceforge.io/`
Stan diagnostics and CmdStanR: `https://mc-stan.org/learn-stan/diagnostics-warnings.html` and `https://mc-stan.org/cmdstanr/`

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